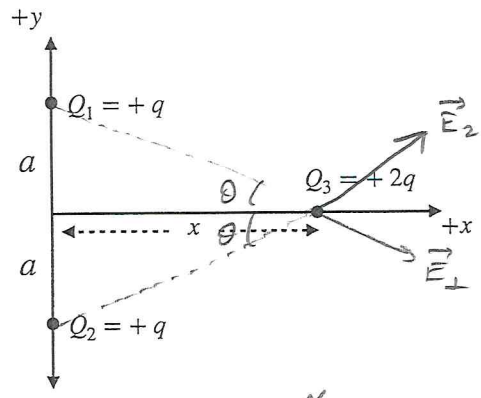
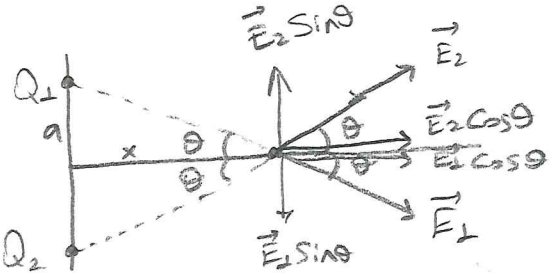
Problem 1 (10 points)

Three point charges are placed on the xy-plane as shown in the figure such that Q_1 is at $(0, a)$, Q_2 is at $(0, -a)$ and Q_3 is at $(x, 0)$. The charges are $Q_1 = Q_2 = +q$, $Q_3 = +2q$.



$$\cos\theta = \frac{x}{\sqrt{x^2+a^2}}$$

(a) (7 pts.) Find the electric field and electrical force on Q_3.



$$|\vec{E}_\perp| = |\vec{E}_2| = \frac{1}{4\pi\epsilon_0}\frac{q}{(x^2+a^2)}$$

$$\vec{E}_\perp \sin\theta + \vec{E}_2 \sin\theta = 0$$

$$\vec{E}_{total} = \vec{E}_\perp \cos\theta + \vec{E}_2 \cos\theta = E_\perp \cos\theta \uparrow + E_2 \cos\theta \uparrow = (E_\perp + E_2)\cos\theta \uparrow$$

$$\vec{E}_{total} = \frac{1}{4\pi\epsilon_0}\frac{2q}{(x^2+a^2)} \cdot \cos\theta \uparrow \Rightarrow \vec{E}_{total} = \frac{1}{4\pi\epsilon_0}\frac{2qx}{(x^2+a^2)^{3/2}}(\uparrow)$$

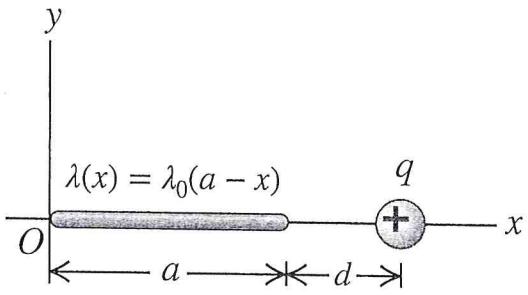
$$\vec{F}_3 = Q_3 \cdot \vec{E}_{total} \Rightarrow \vec{F}_3 = \frac{1}{4\pi\epsilon_0}\frac{4q^2x}{(x^2+a^2)^{3/2}}(\uparrow)$$

(b) (3 pts.) How would the electric field and force on Q_3 change if it's charge is replaced with $-4q$.

- Electric field doesn't change.
- Electric force is doubled and reversed.

Problem 2 (25 points)

Positive charge is distributed along the $x-$axis from $x = 0$ to $x = a$. The distribution as a function of x is given as $\lambda(x) = \lambda_0(a - x)$ where λ_0 is a positive constant with appropriate units. A positive point charge q is located on the positive $x-$axis at $x = a + d$, a distance d to the right of the end of distribution.



(a) (5 pts.) Express the unit of λ_0 in meters (m) and Coulomb (C).

$\lambda = \frac{Q}{L}$ $Q \to [C]$, $L \to [m]$

$\lambda(x) = \lambda_0(a-x)$

$[\frac{C}{m}] = [?]\,[m] \Rightarrow [?] = [\frac{C}{m^2}]$

(b) (10 pts.) Find the total charge of the distribution.

$\lambda = \frac{dq}{dx} \Rightarrow \int dq = \int \lambda\, dx \Rightarrow q = \int_{x=0}^{x=a} \lambda_0(a-x)\,dx$

$q = \lambda_0 \int_{x=0}^{x=a} (a-x)\,dx = \lambda_0 \left[\int_0^a a\,dx - \int_0^a x\,dx \right] = \lambda_0 \left[ax \Big|_0^a - \frac{x^2}{2}\Big|_0^a \right]$

$q = \lambda_0 \left[a^2 - \frac{a^2}{2} \right] = \lambda_0 \frac{a^2}{2}$

(c) (10 pts.) Find the electric force exerted on the charge distribution.

$\lambda = \frac{dq}{dx} \Rightarrow dq = \lambda(x)\,dx$

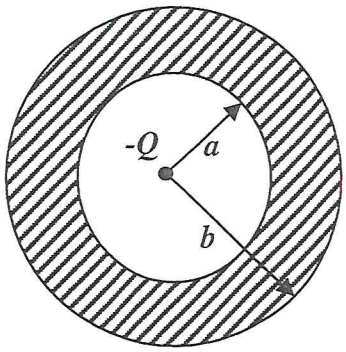
Electric Force exerted on dq by point charge : $dF = dq\,E$

$\int dF = \int dq \frac{1}{4\pi\epsilon_0} \frac{q}{(a+d-x)^2} = \int \frac{1}{4\pi\epsilon_0} \frac{q \cdot \lambda_0 (a-x)\,dx}{(a+d-x)^2} \Rightarrow F = \frac{q\lambda_0}{4\pi\epsilon_0} \int_{x=0}^{x=a} \frac{(a-x)\,dx}{(a+d-x)^2}$

$a - x + d = u$, $\boxed{a - x = u - d}$, $\boxed{-dx = du}$ $\Rightarrow \int \frac{(a-x)\,dx}{(a+d-x)^2} = \int \frac{(u-d) \cdot -du}{u^2} = \dots$

$F = \frac{q\lambda_0}{4\pi\epsilon_0} \cdot -\left\{ \ln\left(\frac{d}{a+d}\right) + \left(1 - \frac{d}{a+d}\right) \right\}$

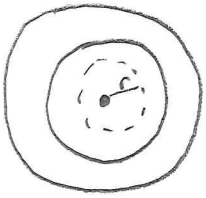
Problem 3 (20 points)



A point charge $-Q$ is placed at the center of a hollow spherical conducting shell with inner radius a and outer radius b as shown in the figure. The total charge of the shell is $+3Q$.

(a) (10 pts.) Find the electric field $\vec{E}(r)$ for $r < a$, $a < r < b$ and $r > b$.

$r < a$



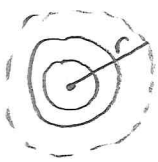
$$\oint \vec{E} \cdot d\vec{A} = \frac{|Q_{enclosed}|}{\epsilon_0} \qquad E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0} \Rightarrow E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

Radially inwards direction $\rightarrow \vec{E}(r) = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}(-\hat{r})$

$a < r < b$

Inside the conducting shell, Electric field is zero. $(\vec{E}(r) = 0)$

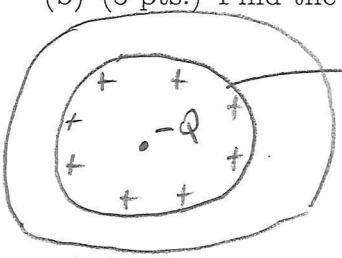
$r > b$



$$E \cdot 4\pi r^2 = \frac{2Q}{\epsilon_0} \Rightarrow E = \frac{1}{4\pi\epsilon_0}\frac{2Q}{r^2}$$

Radially outwards direction $\rightarrow \vec{E}(r) = \frac{1}{4\pi\epsilon_0}\frac{2Q}{r^2}(\hat{r})$

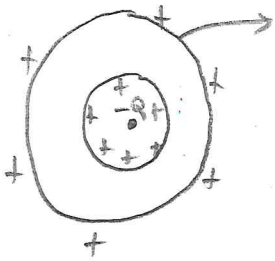
(b) (5 pts.) Find the surface charge density on the inner surface of the shell..



On the inner surface, total charge would be $+Q$.

$$\rho_{inner} = \frac{Q}{4\pi a^2}$$

(c) (5 pts.) Find the surface charge density on the outer surface of the shell.

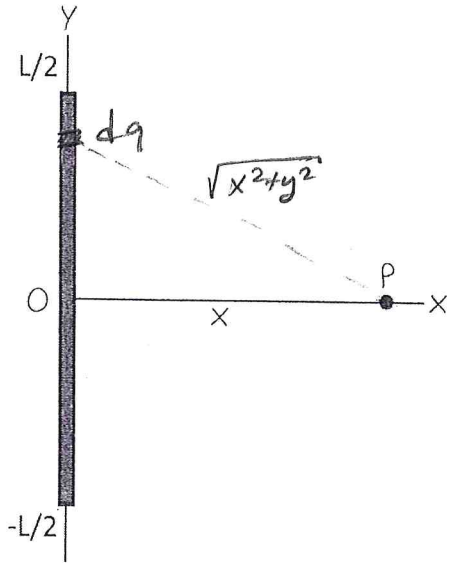


On the outer surface, total charge would be $+2Q$

$$\rho_{outer} = \frac{2Q}{4\pi b^2}$$

Problem 4 (25 points)

Electrical charge is non-uniformly distributed on a rod of length L, which lies on the $y-$axis. The linear charge density of the rod changes with the y coordinate such that $\lambda(y) = \lambda_0 y$ where λ_0 is a positive constant number. The rod is positioned symmetrically between $y = -L/2$ and $y = L/2$.



$$\lambda = \frac{dq}{dy} \Rightarrow dq = \lambda(y)\,dy$$

$$dq = \lambda_0 y\,dy$$

(a) (18 pts.) Calculate the **electrical potential** V_P at a point P which is on the $x-$axis.

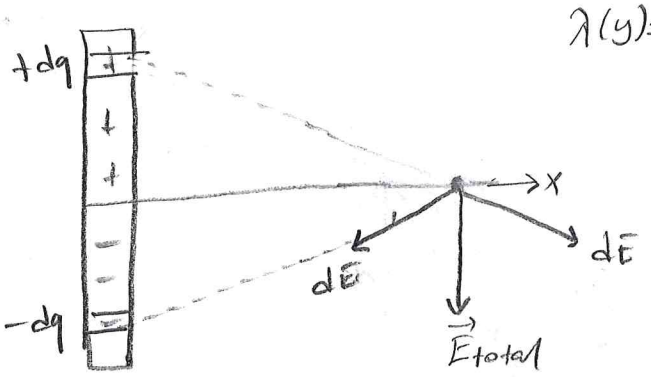
$$V = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r} = \frac{1}{4\pi\epsilon_0}\int \frac{\lambda_0 y\,dy}{(x^2+y^2)^{1/2}} = \frac{1}{4\pi\epsilon_0}\cdot\lambda_0 \int_{y=-\frac{L}{2}}^{y=L/2} \frac{y\,dy}{(x^2+y^2)^{1/2}}$$

$$V = k\lambda_0 \left\{ (x^2+y^2)^{1/2} \Big|_{y=-\frac{L}{2}}^{y=\frac{L}{2}} \right\} = 0$$

$$V = -\int \vec{E}\cdot d\vec{x} = 0 \Rightarrow \vec{E}\cdot d\vec{x} = 0$$

$\Rightarrow \vec{E} \perp d\vec{x}$ or $\vec{E}$ is perpendicular to x axis.

(b) (7 pts.) Determine the direction of the electrical field on the $x-$axis.



$\lambda(y) = \lambda_0 y$ If $y > 0$, $\lambda(y) > 0$ and Q is (+)

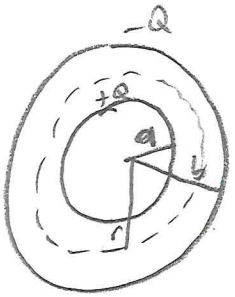
If $y < 0$, $\lambda(y) < 0$ and Q is (−)

There is a symmetrical charge distribution along the y axis. So, total Electric field is on $(-\hat{j})$ direction.

Problem 5 (20 points)

Two spherical conducting shells are placed concentrically. The inner shell has radius a and total charge $+Q$, the outer shell has radius b and total charge $-Q$.

(a) (10 pts.) Find the electric field between the shells.



$$\oint \vec{E}\cdot d\vec{A} = \frac{Q_{enclosed}}{\epsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$$

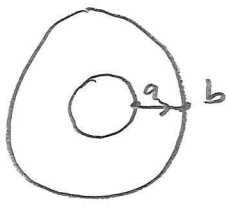
$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}(\hat{r})$$

Because of positively charge source, Electric field is going to outwards direction.

(b) (10 pts.) Find the capacitance of this system.

$$C = \frac{Q}{\Delta V}$$



$$V_a - V_b = \int_a^b \vec{E}\cdot d\vec{l}$$

$$V_a - V_b = \frac{1}{4\pi\epsilon_0}\int_a^b \frac{Q}{r^2}\,dr = \frac{Q}{4\pi\epsilon_0}\int_a^b \frac{dr}{r^2} = \frac{Q}{4\pi\epsilon_0}\left[-\frac{1}{r}\Big|_a^b\right]$$

$$\Delta V = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$$

$$C = \frac{\cancel{Q}}{\frac{\cancel{Q}}{4\pi\epsilon_0}\left(\frac{b-a}{ab}\right)} = \frac{4\pi\epsilon_0 \cdot a \cdot b}{b-a}$$